Step 4 : Test statistic

Since n is large, $np = 560 > 5$ and $n(1 - p) = 440 > 5$, the test statistic under the null hypothesis, is $Z = \dfrac{p - P}{\sqrt{\dfrac{PQ}{n}}}$.

Its sampling distribution under H_0 is the $N(0,1)$ distribution.

Step 5 : Calculation of Test Statistic

The value of Z can be calculated for the sample information from

$$z_0 = \frac{p - p_0}{\sqrt{\dfrac{p_0 q_0}{n}}} \text{ as}$$

$$z_0 = \frac{0.56 - 0.50}{\sqrt{\dfrac{0.5 \times 0.5}{1000}}}$$

Thus, $z_0 = 3.79$

Step 6 : Critical value

Since H_1 is a two-sided alternative hypothesis, the critical value at 1% level of significance is $z_{\alpha/2} = z_{0.005} = 2.58$.

Step 7 : Decision

Since H_1 is a two-sided alternative, elements of the critical region are determined by the rejection rule $|z_0| \geq z_e$. Thus it is a two-tailed test. Since $|z_0| = 3.79 > z_e = 2.58$, reject H_0 at 1% level of significance. Therefore, there is significant evidence to conclude that the preference of tea and coffee are different.

1.14 TEST OF HYPOTHESES FOR EQUALITY OF PROPORTIONS OF TWO POPULATIONS

Procedure:

Step 1 : Let P_X and P_Y denote respectively the proportions of Population-1 and Population-2 possessing the qualitative characteristic (attribute) under study. Frame the null hypothesis as H_0: $P_X = P_Y$ and choose the suitable alternative hypothesis from

(i) H_1: $P_X \neq P_Y$ (ii) H_1: $P_X > P_Y$ (iii) H_1: $P_X < P_Y$

Step 2 : **Let p_X and p_Y denote respectively the proportions of the samples of sizes m and n** drawn from Population-1 and Population-2 possessing the attribute, where m and n are large (*i.e.*, $m \geq 30$ and $n \geq 30$). Also, $mp_X > 5$, $m(1 - p_X) > 5$, $np_Y > 5$ and $n(1 - p_Y) > 5$. Here, these two samples are assumed to be independent.

Step 3 : Specify the level of significance, α.

12th_Statistics_EM_Unit_1.indd 25 1/28/2020 4:47:39 PM